Week 7: Bayesian statistical inference

F79MA Statistical Models A

Heriot-Watt University

Bayesian Inference

Bayes' Theorem, priors, likelihoods and posteriors

Bayesian inference is a comparatively new approach to statistical inference. It is now a major area in statistics and is used in many areas including biology, medicine, engineering, physics and epidemiology. At one time nearly all statisticians were classical (frequentist) and Bayesian methods were little used. However, thanks to the advent of powerful computing resources, Bayesian methods can be applied to many real-world problems that could not easily be tackled using classical methods.

The basic starting point

A key idea in Bayesian statistics is that all quantities about which there is uncertainty should be represented as random variables and characterised using probability distributions that illustrate our knowledge about these quantities.

Let us consider a coin with an 'H' face and an 'F' face which we decide to toss.

If we let $\mathbf{P}(C = `H') = p$ then upon tossing the coin a frequentist would consider the likelihood $\mathbf{P}(C \mid p)$ where C denotes the observed coin face and p is a magical unknown but fixed probability that we would love to know, and which we could estimate by maximising this likelihood.

Bayesians instead would argue that we actually observe C so think we should consider $\mathbf{P}(p \mid C)$. This relates to Bayes' theorem because

$$\mathbf{P}(p \mid C) = \frac{\mathbf{P}(C \mid p)\mathbf{P}(p)}{\mathbf{P}(C)},$$

where $\mathbf{P}(C \mid p)$ is the likelihood and $\mathbf{P}(p)$ is a prior belief about p.

A simple example

Let us illustrate these ideas further with the following 'toy' example:

Suppose we have a bag of coins with two types of coins. The majority of coins (type 1) have a 'H' face and a 'T' face, but a small proportion (1%) have been minted with 2 'H' faces (type 2). It is known that the coins, when tossed, will land one side or the other with equal probability.

I take a coin from the bag and toss it 3 times, obtaining 3 'H' s. How can you analyse the result of this experiment to infer the particular type of coin I have tossed?

Let p denote the probability of getting an 'H' with my coin. Then before tossing the coin you know that

$$p = 0.5 \text{ OR } p = 1.0$$

depending on whether the coin is type 1 or type 2. If you assume that the coin is randomly drawn from the population of coins then

$$\begin{aligned}\mathbf{P}(p = 0.5) &= 0.99\\ \mathbf{P}(p = 1.0) &= 0.01.\end{aligned}$$

For the observed outcome from the experiment you know that

$$\begin{aligned}\mathbf{P}(\text{HHH}|p = 0.5) &= 0.125\\ \mathbf{P}(\text{HHH}|p = 1.0) &= 1.0.\end{aligned}$$

You can now calculate the following conditional probabilities using Bayes' Theorem:

$$\begin{aligned}\mathbf{P}(p = 0.5|\text{HHH}) &= \frac{0.99 \times 0.125}{0.99 \times 0.125 + 0.01 \times 1.0}\\ &= 0.925 \text{ (3 dp)}\end{aligned}$$

Also

$$\mathbf{P}(p = 1.0|\text{HHH}) = 0.075 \text{ (3 dp)}$$

This calculation is an illustration of Bayesian reasoning and the main steps in achieving it are as follows:

(A) You began with an initial (prior) belief about the value of p.

(B) You carried out an experiment.

(C) You computed the conditional distribution of p given the observed outcome 'HHH'. (This is the 'posterior' distribution of p.)

In Bayesian inference, we express our conclusions from an experiment in the form of a probability distribution for the quantity(ies) or parameters of interest. Before the experiment your prior belief was: $\mathbf{P}(\text{Type 1}) = 0.99$, $\mathbf{P}(\text{Type 2}) = 0.01$. Denote this probability function as $\pi(p)$, where $p \in \{0.5, 1\}$. We refer to this as the *prior distribution of p*.

After observing data y = 'HHH' your posterior belief becomes: $\mathbf{P}(\text{Type 1}) = 0.925$, $\mathbf{P}(\text{Type 2}) = 0.075$. We denote this as $\pi(p|y)$ and we call it the *posterior distribution of* p.

How did you go from prior to posterior? Note that the above calculation of the posterior could be written as

$$\pi(p|y) = \frac{\pi(p)\mathbf{P}(y|p)}{\sum_{q\in\{0.5,1\}} \pi(q)\mathbf{P}(y|q)}$$

Note also that $\mathbf{P}(y|p)$ is just our likelihood function $L(p; y)$. You obtain your 'posterior' distribution $\pi(p|y)$ by multiplying the prior $\pi(p)$ by $L(p; y)$, and 'normalising' to turn it into a probability mass function. This, in essence, is how Bayesian inference works and all the calculations we subsequently do are based on this idea.

To a classical statistician, there is nothing contentious in the above example since the value of p could be considered to be generated through a repeatable experiment (random sampling of a coin from the whole population). Bayesian inference represents all unknown parameters of interest as random quantities, even when their value has not been assigned through some repeatable sampling process. This is seen by some statisticians as being controversial as the specification of distributions to unknown parameters potentially introduces subjective belief into the process of statistical inference. Conversely, other statisticians see it as a principled strategy to introduce subjective belief into the process of statistical inference in order to combine it with the information from data.

Estimating parameters using Bayesian inference

Suppose that we carry out an experiment where the outcome y has probability function $f(y; \theta)$ where θ is an unknown parameter of interest. Suppose my prior belief about θ is described by the probability density/mass function $\pi(\theta)$. Then, on observing y, my posterior belief about θ is given using Bayes' theorem as:

$$\pi(\theta|y) = \frac{\pi(\theta)L(\theta; y)}{\int \pi(\theta')L(\theta'; y)d\theta'}, \qquad (**)$$

where $L(\theta; y) = f(y; \theta) = \pi(y|\theta)$ is the likelihood. We have written the above rule for the case where θ is a continuously-valued parameter so that $\pi(\theta)$ is a density. If $\pi(\theta)$ is a discrete distribution – as in the above example with the coin - then the integral on the denominator just becomes a sum. Conceptually, there is no real difference between the situations.

Some things to note:

- Bayesians tend to use $\pi()$ to denote a probability density or mass function – be careful not to get confused with the number π!
- To calculate the posterior distribution you first need to identify a prior $\pi(\theta)$ – this is the main source of controversy in Bayesian statistics – see the next section.

- The only way in which the observations enter Bayesian inferences is through the likelihood function. Therefore Bayesian methods respect the likelihood principle.
- Although the definition of $\pi(\theta|y)$ involves an integral over the whole parameter space (on the denominator), we can often circumvent the computation of this integral (see later).

Example of Bayesian inference.

Example 2 - Bayesian inference for Bin(n,p): You want to estimate the proportion p of the population who intend to vote for the conservative party in a forthcoming council election in your town. To do this you select a random sample of $n = 100$ names from the voting register and contact them to ask their voting intentions. At the last 10 elections, 40% the people in the town voted for the conservative party on average, with a standard deviation of 14.8%. So we take 0.4 as the prior expectation of p, and 0.148^2 as the prior variance of p.

In your survey you find that $y = 25$ people in your sample say they will vote conservative. Carry out a Bayesian analysis using this information.

Clearly the sampling distribution for y is $\text{Bin}(n, p)$ where p is the unknown proportion of conservative voters. We can therefore identify our likelihood as:

$$L(p; y) = \binom{n}{y} p^y (1-p)^{n-y}$$

We need to specify our prior distribution for p. This should be done before the data are collected. Since p is a probability, we need to find a distribution whose range is the interval (0, 1). Now we have prior information from the last election, but this does not fully specify the prior distribution of p. What about using a Beta distribution of some sort, with parameters chosen such that $\mathbf{E}(p) = 0.4$ and $\mathbf{Var}(p) \approx (0.148)^2$? This corresponds to a Beta(4,6) distribution (recall that for a random variable $p \sim \text{Beta}(\alpha, \beta)$ with $\alpha, \beta > 0$, $\mathbf{E}(p) = \alpha/(\alpha + \beta)$ and $\mathbf{Var}(p) = (\alpha\beta)/[(\alpha + \beta)^2(\alpha + \beta + 1)]$).

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> x = ppoints(1000)
> plot(x, dbeta(x, 4, 6), type = "l")
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As expected, this distribution has mean 0.4 but also considerable variation around this mean value – representing uncertainty in the true proportion. Now we are ready to consider the observation $y = 25$ and derive the posterior using the formula (**). Let us check that the posterior distribution is Beta(29,81) taking it step by step.

- **Step 1: Prior** First, we begin with a prior belief $p \sim \text{Beta}(4, 6)$. This means that

$$\pi(p) = \frac{\Gamma(4+6)}{\Gamma(4)\Gamma(6)} p^{4-1}(1-p)^{6-1} = \frac{\Gamma(10)}{\Gamma(4)\Gamma(6)} p^3(1-p)^5.$$

- **Step 2: Data** Here $Y \sim \text{Bin}(n, p)$ where we observe $Y = 25$ from a sample with $n = 100$. This gives a likelihood of

$$L(p; Y = 25) = \binom{100}{25} p^{25}(1-p)^{100-25} = \binom{100}{25} p^{25}(1-p)^{75}.$$

- **Step 3: Posterior** Using (**) we have

$$\begin{aligned}
\pi(p|y) &= \frac{\pi(p)L(p; Y = 25)}{\int \pi(q)L(q; Y = 25)dq} \\
&= \frac{\frac{\Gamma(10)}{\Gamma(4)\Gamma(6)} p^3(1-p)^5 \binom{100}{25} p^{25}(1-p)^{75}}{\int_{q=0}^1 \frac{\Gamma(10)}{\Gamma(4)\Gamma(6)} q^3(1-q)^5 \binom{100}{25} q^{25}(1-q)^{75} dq} \\
&= \frac{p^{28}(1-p)^{80}}{\int_{q=0}^1 q^{28}(1-q)^{80} dq} \\
&= \frac{\frac{\Gamma(110)}{\Gamma(29)\Gamma(81)} p^{28}(1-p)^{80}}{\int_{q=0}^1 \frac{\Gamma(110)}{\Gamma(29)\Gamma(81)} q^{28}(1-q)^{80} dq} \\
&= \frac{\Gamma(110)}{\Gamma(29)\Gamma(81)} p^{28}(1-p)^{80}
\end{aligned}$$

  It follows that the posterior distribution of p given $Y = 25$ is Beta(29,81).

In the above we can see that we had various constants that disappeared in our derivation. We also know that if $\pi(p|y)$ is the density of a continuous random variable taking values between 0 and 1 then we must have $\int_{p=0}^1 \pi(p|y)dp = 1$. It follows that we can make the above step-by-step derivation shorter:

- **Step 1: Prior** First, we begin with a prior belief $p \sim \text{Beta}(4, 6)$. This means that

$$\pi(p) \propto p^{4-1}(1-p)^{6-1} = p^3(1-p)^5.$$

- **Step 2: Data** Here $Y \sim \text{Bin}(n, p)$ where we observe $Y = 25$ from a sample with $n = 100$. This gives a likelihood of

$$L(p; Y = 25) \propto p^{25}(1-p)^{100-25} = p^{25}(1-p)^{75}.$$

- **Step 3: Posterior** Up to proportionality we have

$$\pi(p|y) \propto \pi(p)L(p; Y = 25) \propto p^3(1-p)^5 p^{25}(1-p)^{75} = p^{28}(1-p)^{80}.$$

  In other words, $\pi(p|y) = Cp^{28}(1-p)^{80}$ for some constant C.

Since $\int_{p=0}^{1} \pi(p|y)dp = 1$, we have

$$\int_{p=0}^{1} Cp^{28}(1-p)^{80}dp = 1$$

and hence

$$C = \frac{1}{\int_{p=0}^{1} p^{28}(1-p)^{80}dp} = \frac{\Gamma(110)}{\Gamma(29)\Gamma(81)},$$

giving the required result. Alternatively, without calculating the integral, we know that a Beta(29,81) distribution has a density proportional to $p^{28}(1-p)^{80}$ and so using the constraint that

$$\int_{p=0}^{1} Cp^{28}(1-p)^{80}dp = 1$$

implies that C must be the coefficient in a Beta(29,81) density.

Note that all of the above could be done in complete generality for data y from $\text{Bin}(n,p)$ and prior with $p \sim \text{Beta}(\alpha,\beta)$. In this case one can show (try!) that the posterior distribution would be $\text{Beta}(\alpha+y, \beta+n-y)$.

Exercise. Plot the posterior density function for Beta(29,81) and compare it with the density function of the prior Beta(4, 6). These plots illustrate the way in which our belief changes in response to the observation $y = 25$.

Let us now compare our result with the classical approach to point estimation. From a classical perspective, we would ignore past data (i.e. we would not have a prior belief about p). Instead, if we let $Y \sim \text{Bin}(n,p)$ and observe $Y = y = 25$ for $n = 100$, our maximum likelihood estimate for p would be

$$\hat{p} = \frac{y}{n} = \frac{25}{100} = 0.25.$$

On the other hand, we found with a prior distribution of Beta(4, 6) we obtained a posterior distribution for p of Beta(29,81). If we now use the mean of this posterior distribution as an estimate for p instead of using maximum likelihood estimation, we find that

$$\hat{p} = (p|Y = 25) = \frac{29}{29+81} = 0.2639 > 0.25.$$

Why is this bigger? Our prior belief was that $p = 0.4$. Hence this prior belief would lead to a higher estimate of p given our data.

Example 3 - Bayesian inference for Exp(λ): Suppose we have a population where lifetimes are known to be exponentially distributed with parameter λ. We collect a random sample of n lifetimes $y = (x_1, \ldots, x_n)$. Carry out inference on λ using Bayesian methods.

The main steps are as follows:

- **Step 1:** We need to specify a prior distribution for λ representing belief about λ before any observations are taken. Given λ is positive, which family of distributions might be sensible to consider? What about Gamma(α, β)?

  Set $\pi(\lambda)$ as Gamma(α, β) where α, β are chosen to reflect your prior belief about λ (for now, we keep them unspecified).

- **Step 2:** Note that the likelihood is:

$$L(\lambda, y) = \lambda^n \exp\left\{-\lambda \sum_{i=1}^{n} x_i\right\}$$

- **Step 3:** By using Bayes' theorem (the formula (**)), we find that the posterior distribution is Gamma$(\alpha + n, \beta + \sum x_i)$.

Exercise 1 - Bayesian inference for Poisson(λ): Suppose now we have a random sample $y = (x_1, \ldots, x_n)$ of size n from a Poisson(λ) distribution where our prior belief regarding λ is represented as a Gamma(α, β) distribution. Using Bayes' theorem, show that $\lambda|y$ is a Gamma$(\alpha + \sum x_i, \beta + n)$ random variable.

Exercise 2 - Bayesian inference for N(μ, σ^2) distribution where σ^2 is known: Let $y = (x_1, \ldots, x_n)$ be a random sample from a N(μ, σ^2) distribution where σ^2 is known. Suppose we use a N(α, β) prior for μ. Use Bayes' theorem to derive the posterior distribution $\pi(\mu|y)$ and comment on the form of posterior expectation of μ.

Conjugacy: In the above examples, the prior distribution and posterior distributions come from the same family. Identifying the posterior distribution was particularly easy in general, and we observed that the likelihood and the prior had the same algebraic form. In these circumstances we say that the prior in question is the *conjugate* prior for the parameter/distribution.

- Example 2: The Beta distribution is the conjugate prior for p in Bin(n, p).
- Example 3: The Gamma distribution is the conjugate prior for λ in Exp(λ).
- Exercise 1: The Gamma distribution is the conjugate prior for λ in Poi(λ).
- Exercise 2: The Normal distribution is the conjugate prior for μ in N(μ, σ^2) where σ^2 is known.